

MENA National Energy Rebound Analysis

Estimating National Energy Rebound Effects using Brockway et al. (2017)

Countries	UAE · Egypt · Jordan · Tunisia · Morocco
Time Period	1990 – 2023 (34 years per country)
Methods	Method 1: AES/PES Method 2: EEE (Brockway et al. 2017)
Production Function	Nested KL-E CES (3-factor, constant-returns)
Base Year	1990
Cost Shares	sE = 0.08 sL = 0.46 sK = 0.46
Data Source	PWT 11.0 (K, L, GDP) OAPEC (Energy in Mtoe)
Date	June 2026

Rebound State Classification

Rebound State	Criterion	Interpretation
Super-conservation	$Re < 0\%$	Energy use falls faster than efficiency gains
Partial rebound	$0\% < Re < 100\%$	Some of efficiency gain is taken back
Full rebound	$Re = 100\%$	All efficiency gains are taken back
Backfire	$Re > 100\%$	Energy use rises beyond original level

1. Introduction

Energy efficiency improvements are often expected to reduce total energy consumption. However, the **energy rebound effect** describes the phenomenon by which some — or all — of the anticipated savings are offset by behavioural, economic, or structural responses. When the rebound exceeds 100%, a *backfire* is said to occur: energy use actually increases beyond the pre-efficiency baseline.

This report estimates national-level energy rebound effects for five MENA economies — **UAE, Egypt, Jordan, Tunisia, and Morocco** — over 1990–2023, using the two complementary macroeconomic methods proposed by **Brockway et al. (2017)**: the AES/PES method (Method 1) and the Energy-Economy-Environment (EEE) method (Method 2). Both methods require estimation of a nested constant-elasticity-of-substitution (CES) production function.

2. Data & Assumptions

Dataset: Sheet1 of the supplied Excel workbook; 170 usable observations after removing header artefacts.

- Energy (Mtoe/yr) used as proxy for *both* primary energy and useful exergy — no separate exergy dataset available.
- All variables normalised to 1990 = 1 as base year.
- $sE = 0.08$ applied uniformly (consistent with Brockway et al. 2017 for developing economies).
- Labour income share = 0.50 $\rightarrow sL = sK = 0.50 \times (1 - sE) = 0.46$.
- λ (embodied technical change) treated as time-invariant for Method 1 annual rebound calculation.

Cost Share Parameters

Parameter	Formula	Value	Verify
sE	Given	0.08	—
sL	$0.50 \times (1 - sE)$	0.46	—
sK	$0.50 \times (1 - sE)$	0.46	—
Sum	$sE + sL + sK$	1.00	✓

3. Production Function

A **nested KL-E constant-elasticity-of-substitution (CES) production function** is fitted separately for each country using non-linear least squares:

$$y_t = \theta \cdot \exp(\lambda \cdot t) \cdot [\delta \cdot (\delta \cdot k_t^{-\rho} + (1-\delta) \cdot l_t^{-\rho})^{\rho/\rho} + (1-\delta) \cdot u_t^{-\rho}]^{-1/\rho}$$

where all lower-case variables are normalised to 1990, and t is years since 1990. Parameters: θ (TFP scale), λ (embodied technical change), δ (K-share in KL nest), δ (KL-share vs Energy), ρ (KL-Energy substitution), ρ (K-L substitution). Elasticities: $\sigma = 1/(1+\rho)$, $\sigma = 1/(1+\rho)$.

Fitting used **50 random starting points** per country to avoid local minima; the solution with lowest RSS was retained. **1000-sample residual bootstrap** provides 95% CIs.

4. Data Overview: Normalized Inputs

Normalized Production Function Inputs (Base Year = 1990)

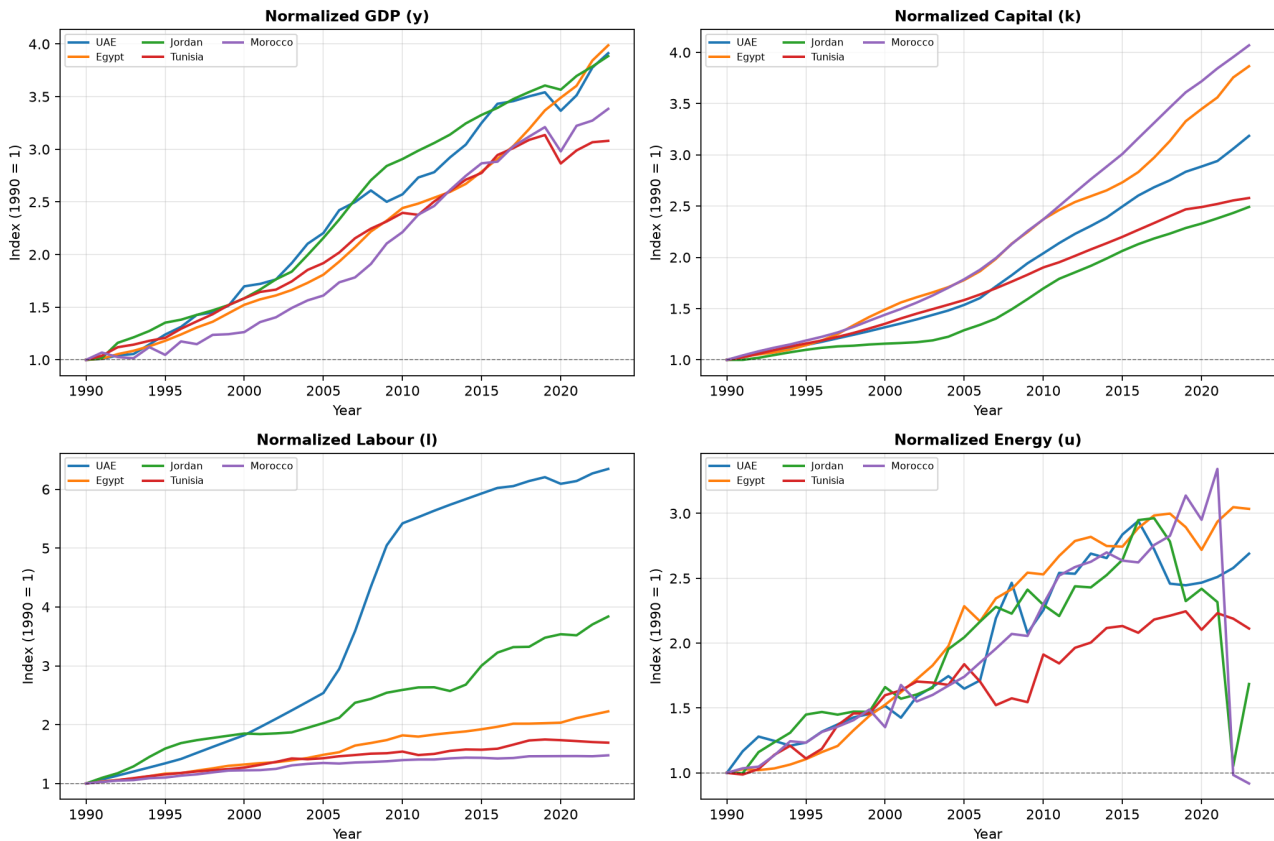


Figure 1. Normalized production function inputs (base year 1990 = 1). Strong growth in GDP (y) and Capital (k) is visible in all countries, particularly UAE. Labour (l) diverges notably: UAE exhibits rapid workforce expansion, while Tunisian and Moroccan labour inputs remain relatively stable.

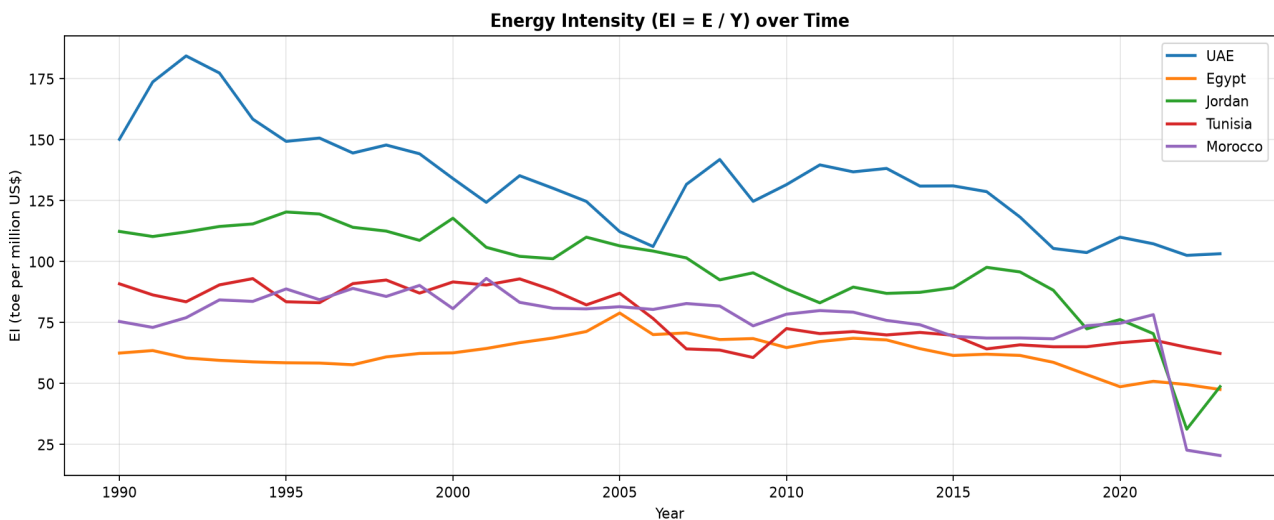


Figure 2. Energy intensity (EI = E/Y, in toe per million US\$) over time. A general declining trend indicates improving energy efficiency in all five economies.

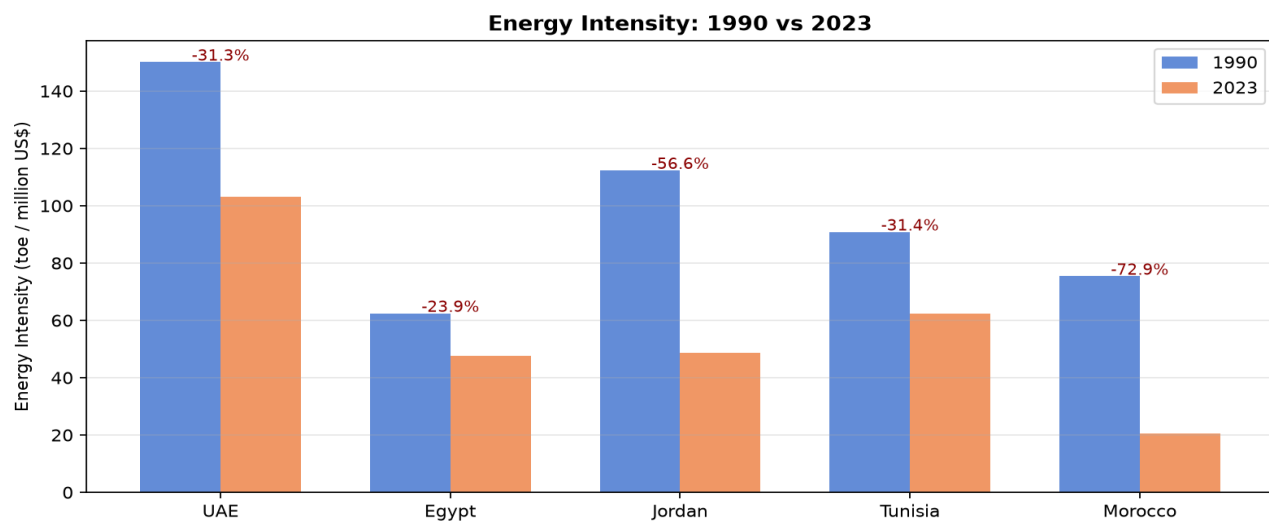


Figure 3. Energy intensity comparison 1990 vs 2023. Percentage change annotated above each pair.

5. CES Production Function Results

Table 1 reports the best-fit CES parameters and 95% bootstrap confidence intervals for each country. R^2 values above 0.97 confirm excellent model fit across all five economies.

Country	θ [CI]	λ [CI]	δ	$\delta_{\square}$	ρ [CI]	σ (KL-E)	$\sigma_{\square}$ (K-L)	R^2
UAE	1.0240 [0.972, 1.119]	-0.0046 [-0.008, 0.002]	0.1514	0.9842	5.0000 [-0.500, 5.000]	0.1667	0.1667	0.9832
Egypt	0.9974 [0.982, 1.015]	0.0043 [0.001, 0.007]	0.7573	0.9990	-0.5000 [-0.500, 5.000]	2.0000	2.0000	0.9993
Jordan	1.0313 [0.905, 1.159]	0.0089 [0.006, 0.012]	0.4556	0.7864	-0.5000 [-0.500, 2.584]	2.0000	2.0000	0.9733
Tunisia	1.0668 [1.004, 1.122]	0.0092 [0.005, 0.016]	0.8471	0.8621	2.7590 [-0.500, 5.000]	0.2660	0.1667	0.9837
Morocco	0.9104 [0.851, 0.951]	0.0025 [-0.000, 0.012]	0.9913	0.9928	-0.5000 [-0.500, 4.513]	2.0000	0.1667	0.9928

Table 1. CES parameter estimates. CI = 95% bootstrap confidence interval. ρ at bound (-0.5 or 5.0) indicates the optimizer reached the parameter boundary.

5.1 Elasticity of Substitution Interpretation

Country	σ (KL-E)	Interpretation
UAE	0.167	KL & Energy are complements ($\sigma < 1$) — limited substitutability
Egypt	2.000	KL & Energy are strong substitutes ($\sigma > 1$, at boundary)
Jordan	2.000	KL & Energy are strong substitutes ($\sigma > 1$, at boundary)
Tunisia	0.266	KL & Energy are weak complements ($\sigma < 1$)
Morocco	2.000	KL & Energy are strong substitutes ($\sigma > 1$, at boundary)

Table 2. KL-Energy elasticity of substitution and economic interpretation.

Note: Egypt, Jordan, and Morocco converge to the lower bound of ρ ($-0.5 \rightarrow \sigma = 2.0$), indicating the optimizer could not identify an interior solution. This is a known challenge with short time-series CES estimation and should be interpreted with caution. Wide bootstrap intervals for ρ in these countries confirm high parameter uncertainty.

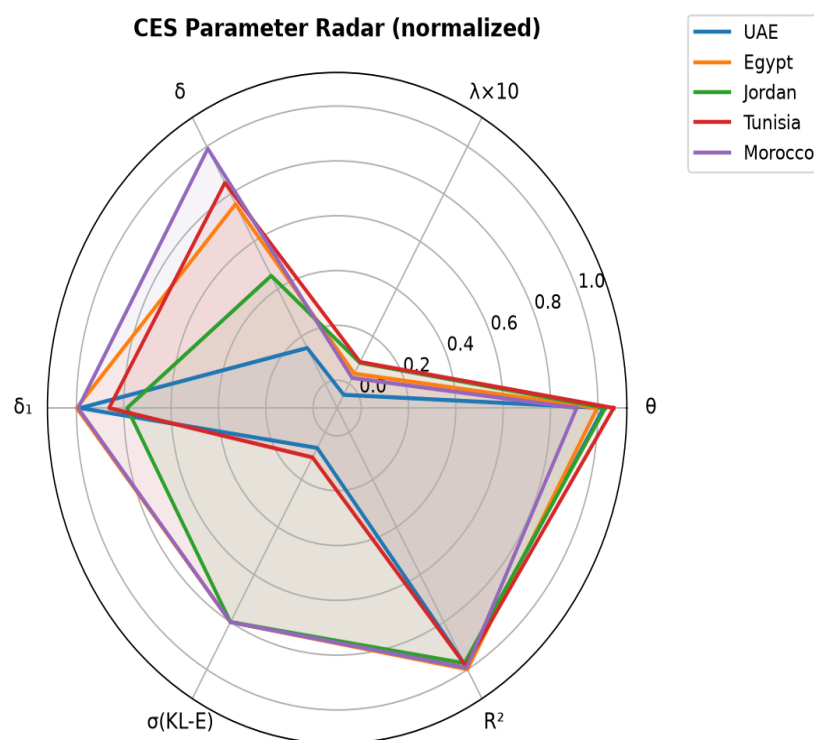


Figure 4. Radar chart of normalised CES parameters across countries (σ capped at 2.5 for scaling).

6. Method 1: AES/PES Rebound

Equations:

Potential Energy Saved	$PES_{\text{country}} = Y_{\text{country}} \times (EI_{\text{country}} - EI_{\text{base}})$
Rebound Energy Consumed	$(PES - AES)_{\text{country}} = \lambda \times (Y_{\text{country}} - Y_{\text{base}}) \times EI_{\text{country}}$
Annual Rebound	$Re_{\text{country}} = [(PES - AES) / PES] \times 100$
Mean Rebound (Method 1)	$Re_M1 = (1/N) \times \sum Re_{\text{country}} \text{ [valid years only, } PES > 0]$
Decomposition	$Re_{\text{country}} = \lambda \times [(Y_{\text{country}} - Y_{\text{base}}) / Y_{\text{country}}] \times [EI_{\text{country}} / (EI_{\text{country}} - EI_{\text{base}})] \times 100$

Years with $PES \leq 0$ (energy intensity did not decline) are excluded from the mean; this reflects periods when technological progress did not improve energy efficiency.

Country	Valid Years	Re_M1 (2.5%)	Re_M1 (base)	Re_M1 (97.5%)	State
UAE	20	-0.90%	-0.50%	0.25%	Super-conservation
Egypt	18	0.22%	1.40%	2.13%	Partial rebound
Jordan	20	0.73%	1.18%	1.57%	Partial rebound
Tunisia	19	0.91%	1.77%	3.00%	Partial rebound
Morocco	19	-0.15%	0.82%	3.83%	Partial rebound

Table 3. Method 1 results summary with 95% bootstrap CI (derived from λ uncertainty).

Method 1 (AES/PES): Annual Rebound per Country

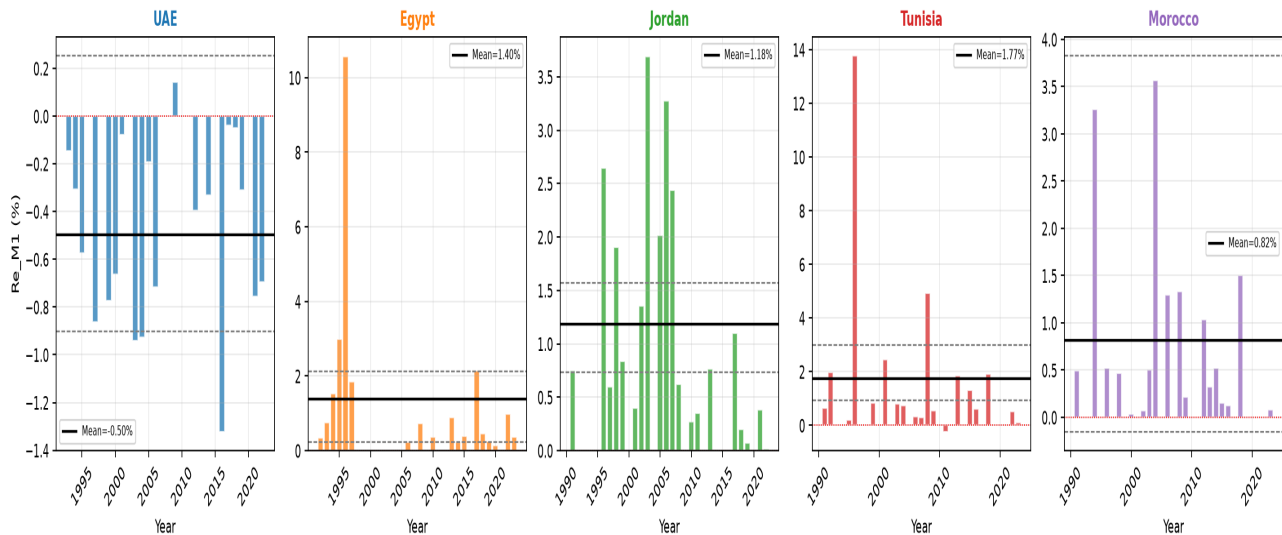


Figure 5. Annual Method 1 rebound per country. Bars show valid years only. Horizontal lines: solid = mean base-fit, dashed = 95% bootstrap CI bounds.

6.1 Annual Rebound Decomposition Tables

For each country, the table below shows all valid years (PES > 0) with the full $A \times B \times C$ decomposition of annual rebound.

UAE ($\lambda = -0.004599$)

Year	PES (Mtoe)	(PES-AES)(Mtoe)	Re annual (%)	$A=\lambda$	$B=(\Delta Y/Y)$	$C=(EI\blacksquare/\Delta EI)$
1993	1.4965	-0.002170	-0.1450	-0.004599	0.012454	25.3188
1994	4.3894	-0.013390	-0.3051	-0.004599	0.079209	8.3747
1995	2.2881	-0.013094	-0.5723	-0.004599	0.075943	16.3861
1997	1.7694	-0.015284	-0.8638	-0.004599	0.079621	23.5909
1999	1.1011	-0.008536	-0.7752	-0.004599	0.042009	40.1276
2000	3.4885	-0.023078	-0.6615	-0.004599	0.108867	13.2138
2001	3.4051	-0.002704	-0.0794	-0.004599	0.013570	12.7235
2003	1.9970	-0.018797	-0.9413	-0.004599	0.080887	25.3043
2004	2.3003	-0.021308	-0.9263	-0.004599	0.087313	23.0701
2005	5.5335	-0.010670	-0.1928	-0.004599	0.046303	9.0558
2006	2.9875	-0.021438	-0.7176	-0.004599	0.089562	17.4235
2009	8.6923	0.012485	0.1436	-0.004599	-0.042997	7.2643
2012	1.6073	-0.006348	-0.3950	-0.004599	0.017918	47.9344
2014	4.4592	-0.014844	-0.3329	-0.004599	0.039991	18.1008
2016	1.6376	-0.021661	-1.3228	-0.004599	0.052685	54.5962
2017	7.2902	-0.002778	-0.0381	-0.004599	0.007297	11.3539
2018	9.1365	-0.004456	-0.0488	-0.004599	0.012969	8.1785
2019	1.2062	-0.003748	-0.3107	-0.004599	0.010962	61.6362
2021	1.9362	-0.014640	-0.7561	-0.004599	0.041730	39.3990
2022	3.6208	-0.025182	-0.6955	-0.004599	0.069851	21.6508

Egypt ($\lambda = 0.004321$)

Year	PES (Mtoe)	(PES-AES)(Mtoe)	Re annual (%)	$A=\lambda$	$B=(\Delta Y/Y)$	$C=(EI\blacksquare/\Delta EI)$
1992	1.5916	0.005602	0.3520	0.004321	0.040872	19.9312
1993	0.5289	0.004042	0.7642	0.004321	0.029103	60.7756
1994	0.3546	0.005453	1.5378	0.004321	0.038168	93.2505
1995	0.2174	0.006502	2.9905	0.004321	0.043795	158.0370
1996	0.0717	0.007571	10.5574	0.004321	0.048609	502.6668
1997	0.4451	0.008195	1.8410	0.004321	0.050549	84.2897
2006	8.5009	0.018692	0.2199	0.004321	0.064210	7.9254
2008	2.9969	0.021680	0.7234	0.004321	0.066827	25.0539

2010	4.4756	0.016641	0.3718	0.004321	0.048981	17.5684
2013	0.9074	0.008071	0.8895	0.004321	0.021322	96.5503
2014	4.7495	0.010513	0.2214	0.004321	0.028481	17.9875
2015	3.8685	0.015390	0.3978	0.004321	0.041765	22.0462
2017	0.7489	0.016116	2.1518	0.004321	0.040217	123.8317
2018	4.4987	0.020369	0.4528	0.004321	0.050589	20.7135
2019	8.4197	0.020425	0.2426	0.004321	0.052562	10.6814
2020	8.6481	0.012602	0.1457	0.004321	0.034516	9.7710
2022	2.5762	0.025538	0.9913	0.004321	0.062393	36.7707
2023	3.9769	0.014763	0.3712	0.004321	0.036228	23.7154

Jordan ($\lambda = 0.008950$)

Year	PES (Mtoe)	(PES-AES)(Mtoe)	Re annual (%)	A= λ	B=($\Delta Y/Y$)	C=(EI■/ ΔEI)
1991	0.0620	0.000465	0.7508	0.008950	0.015831	52.9945
1996	0.0334	0.000884	2.6436	0.008950	0.020406	144.7552
1997	0.2283	0.001369	0.5997	0.008950	0.032073	20.8939
1998	0.0667	0.001269	1.9018	0.008950	0.029240	72.6754
1999	0.1697	0.001421	0.8374	0.008950	0.032785	28.5390
2001	0.5833	0.002319	0.3975	0.008950	0.050061	8.8725
2002	0.1913	0.002585	1.3512	0.008950	0.054676	27.6147
2003	0.0529	0.001949	3.6865	0.008950	0.039954	103.0975
2005	0.2252	0.004539	2.0160	0.008950	0.075329	29.9043
2006	0.1459	0.004779	3.2754	0.008950	0.074870	48.8829
2007	0.2081	0.005078	2.4399	0.008950	0.075582	36.0703
2008	0.7140	0.004421	0.6191	0.008950	0.067342	10.2726
2010	0.5694	0.001530	0.2688	0.008950	0.022623	13.2743
2011	0.4919	0.001734	0.3526	0.008950	0.026642	14.7876
2013	0.2376	0.001821	0.7667	0.008950	0.025436	33.6803
2017	0.1912	0.002108	1.1028	0.008950	0.024138	51.0473
2018	0.7803	0.001544	0.1979	0.008950	0.018830	11.7460
2019	1.6725	0.001179	0.0705	0.008950	0.017213	4.5757
2021	0.6282	0.002408	0.3833	0.008950	0.035267	12.1441
2022	4.3452	0.000736	0.0169	0.008950	0.023705	0.7983

Tunisia ($\lambda = 0.009225$)

Year	PES (Mtoe)	(PES-AES)(Mtoe)	Re annual (%)	A= λ	B=($\Delta Y/Y$)	C=(EI■/ ΔEI)
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1991	0.2406	0.001582	0.6577	0.009225	0.037569	18.9792
1992	0.1604	0.003181	1.9828	0.009225	0.072406	29.6864
1995	0.5843	0.001088	0.1862	0.009225	0.022946	8.7947
1996	0.0245	0.003375	13.7644	0.009225	0.066743	223.5649
1999	0.4148	0.003527	0.8502	0.009225	0.056756	16.2387
2001	0.1047	0.002554	2.4389	0.009225	0.036574	72.2869
2003	0.4097	0.003250	0.7932	0.009225	0.044912	19.1462
2004	0.5701	0.004203	0.7373	0.009225	0.058698	13.6161
2006	1.0647	0.003622	0.3402	0.009225	0.049828	7.4017
2007	1.3688	0.004084	0.2984	0.009225	0.062877	5.1440
2008	0.0554	0.002732	4.9324	0.009225	0.040655	131.5227
2009	0.3577	0.001948	0.5445	0.009225	0.029536	19.9833
2011	0.2497	-0.000603	-0.2416	0.009225	-0.007668	34.1594
2013	0.1806	0.003317	1.8370	0.009225	0.038778	51.3535
2015	0.1610	0.002097	1.3023	0.009225	0.023046	61.2594
2016	0.8423	0.005109	0.6065	0.009225	0.057560	11.4231
2018	0.1246	0.002384	1.9138	0.009225	0.025258	82.1370
2022	0.4622	0.002370	0.5129	0.009225	0.025379	21.9065
2023	0.3962	0.000380	0.0960	0.009225	0.004219	24.6628

Morocco ($\lambda = 0.002517$)

Year	PES (Mtoe)	(PES-AES)(Mtoe)	Re annual (%)	A= λ	B=($\Delta Y/Y$)	C=($EI_{\square}/\Delta EI$)
1991	0.2630	0.001291	0.4909	0.002517	0.064977	30.0135
1994	0.0688	0.002241	3.2577	0.002517	0.093858	137.8828
1996	0.5238	0.002748	0.5247	0.002517	0.108871	19.1451
1998	0.4117	0.001924	0.4673	0.002517	0.071279	26.0456
2000	1.2178	0.000407	0.0334	0.002517	0.015676	8.4678
2002	1.3943	0.000955	0.0685	0.002517	0.032096	8.4810
2003	0.3640	0.001826	0.5017	0.002517	0.059416	33.5402
2004	0.0412	0.001471	3.5692	0.002517	0.045818	309.4542
2006	0.1974	0.002558	1.2961	0.002517	0.072011	71.5012
2008	0.1988	0.002645	1.3304	0.002517	0.066516	79.4548
2009	1.7276	0.003654	0.2115	0.002517	0.092587	9.0752
2012	0.1610	0.001661	1.0312	0.002517	0.033437	122.5140
2013	0.8959	0.002906	0.3243	0.002517	0.057604	22.3672

2014	0.4914	0.002558	0.5205	0.002517	0.049360	41.8918
2015	1.3764	0.002107	0.1531	0.002517	0.041637	14.6067
2016	0.2078	0.000261	0.1256	0.002517	0.005185	96.2641
2018	0.1075	0.001614	1.5020	0.002517	0.029744	200.5999
2022	18.3824	0.000284	0.0015	0.002517	0.015042	0.4082
2023	0.7574	0.000580	0.0766	0.002517	0.032913	9.2449

7. Method 2: EEE Rebound

Equation (Brockway et al. 2017, eq. 11):

$$Re_M2 = \{ [(1 + sE + sK)(1+\rho) + \rho(sE - sK - 1) + sE] / [(1 + sE + sK)(1+\rho)] \} \times 100$$

Decomposition into substitution and output effects:

Substitution effect	$\eta_{sub} = \rho(sE - 1) / [(1 + sE + sK)(1 + \rho)]$; $Re_{sub} = (1 + \eta_{sub}) \times 100$
Total effect	$\eta_{total} = Re_M2/100 - 1$
Output effect	$\eta_{output} = \eta_{total} - \eta_{sub}$; $Re_{output} = \eta_{output} \times 100$

Country	ρ	$\sigma(KL-E)$	η_{sub}	Re_sub(%)	η_{out}	Re_out(%)	Re_M2 base	State
UAE	5.0000	0.1667	-0.4978	50.22%	-0.2403	-24.03%	26.19%	Partial rebound
Egypt	-0.5000	2.0000	0.5974	159.74%	0.4026	40.26%	200.00%	Backfire
Jordan	-0.5000	2.0000	0.5974	159.74%	0.4026	40.26%	200.00%	Backfire
Tunisia	2.7590	0.2660	-0.4385	56.15%	-0.2054	-20.54%	35.61%	Partial rebound
Morocco	-0.5000	2.0000	0.5974	159.74%	0.4026	40.26%	200.00%	Backfire

Table 4. Method 2 results with effect decomposition. Re_M2 = 200% indicates backfire (ρ at lower bound $-0.5 \rightarrow \sigma = 2.0$).

Method 2 Bootstrap Confidence Intervals (95%)

Country	Re_M2 (2.5%)	Re_M2 (base)	Re_M2 (97.5%)
UAE	200.00%	26.19%	26.19%
Egypt	200.00%	200.00%	26.19%
Jordan	200.00%	200.00%	36.84%
Tunisia	200.00%	35.61%	26.19%
Morocco	200.00%	200.00%	27.59%

Table 5. Re_M2 bootstrap CI using ρ at 2.5th and 97.5th percentile. Wide intervals (including both 200% and ~26%) reflect high ρ uncertainty.

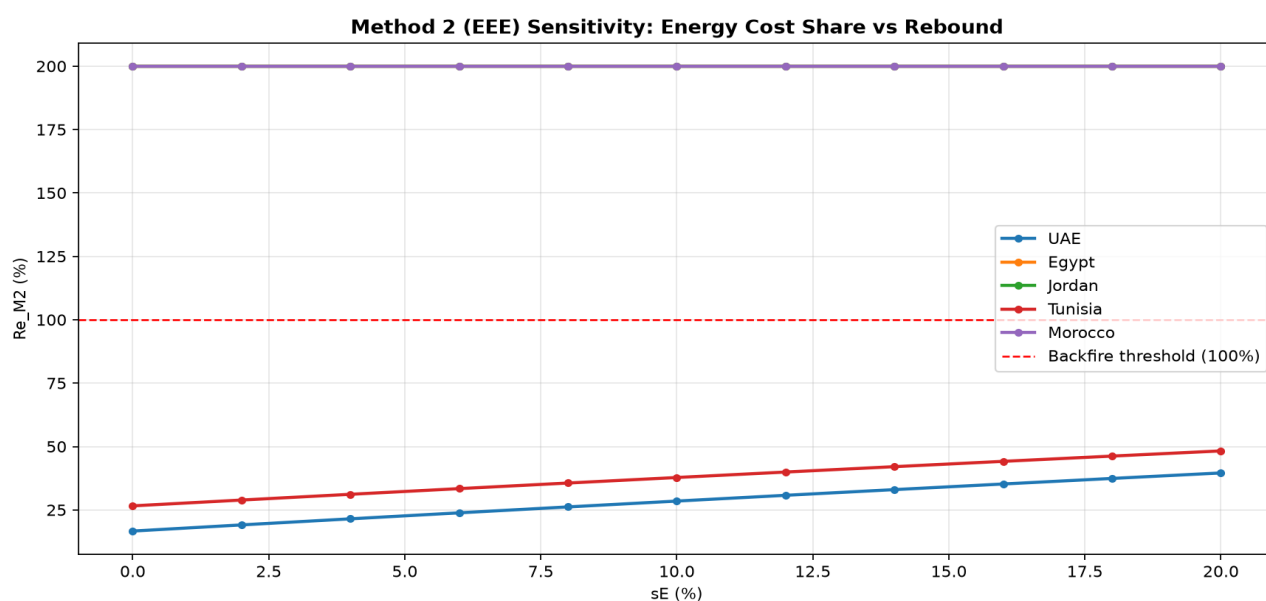


Figure 6. Sensitivity of Re_M2 to energy cost share sE. Countries with ρ at lower bound (-0.5) remain at 200% regardless of sE. Red dashed line marks the backfire threshold.

7.1 Sensitivity Tables: sE vs Re_M2

UAE ($\rho = 5.0000$)

sE (%)	sK (%)	Re_M2 (%)
0%	50.0%	16.67%
2%	49.0%	19.09%
4%	48.0%	21.49%
6%	47.0%	23.86%
8%	46.0%	26.19%
10%	45.0%	28.49%
12%	44.0%	30.77%
14%	43.0%	33.01%
16%	42.0%	35.23%
18%	41.0%	37.42%
20%	40.0%	39.58%

Egypt ($\rho = -0.5000$)

sE (%)	sK (%)	Re_M2 (%)
0%	50.0%	200.00%
2%	49.0%	200.00%
4%	48.0%	200.00%
6%	47.0%	200.00%
8%	46.0%	200.00%

10%	45.0%	200.00%
12%	44.0%	200.00%
14%	43.0%	200.00%
16%	42.0%	200.00%
18%	41.0%	200.00%
20%	40.0%	200.00%

Jordan ($\rho = -0.5000$)

sE (%)	sK (%)	Re_M2 (%)
0%	50.0%	200.00%
2%	49.0%	200.00%
4%	48.0%	200.00%
6%	47.0%	200.00%
8%	46.0%	200.00%
10%	45.0%	200.00%
12%	44.0%	200.00%
14%	43.0%	200.00%
16%	42.0%	200.00%
18%	41.0%	200.00%
20%	40.0%	200.00%

Tunisia ($\rho = 2.7590$)

sE (%)	sK (%)	Re_M2 (%)
0%	50.0%	26.60%
2%	49.0%	28.90%
4%	48.0%	31.17%
6%	47.0%	33.40%
8%	46.0%	35.61%
10%	45.0%	37.79%
12%	44.0%	39.94%
14%	43.0%	42.07%
16%	42.0%	44.16%
18%	41.0%	46.23%
20%	40.0%	48.28%

Morocco ($\rho = -0.5000$)

sE (%)	sK (%)	Re_M2 (%)
0%	50.0%	200.00%
2%	49.0%	200.00%
4%	48.0%	200.00%
6%	47.0%	200.00%
8%	46.0%	200.00%
10%	45.0%	200.00%
12%	44.0%	200.00%
14%	43.0%	200.00%
16%	42.0%	200.00%
18%	41.0%	200.00%
20%	40.0%	200.00%

8. Cross-Country Comparison & Final Summary

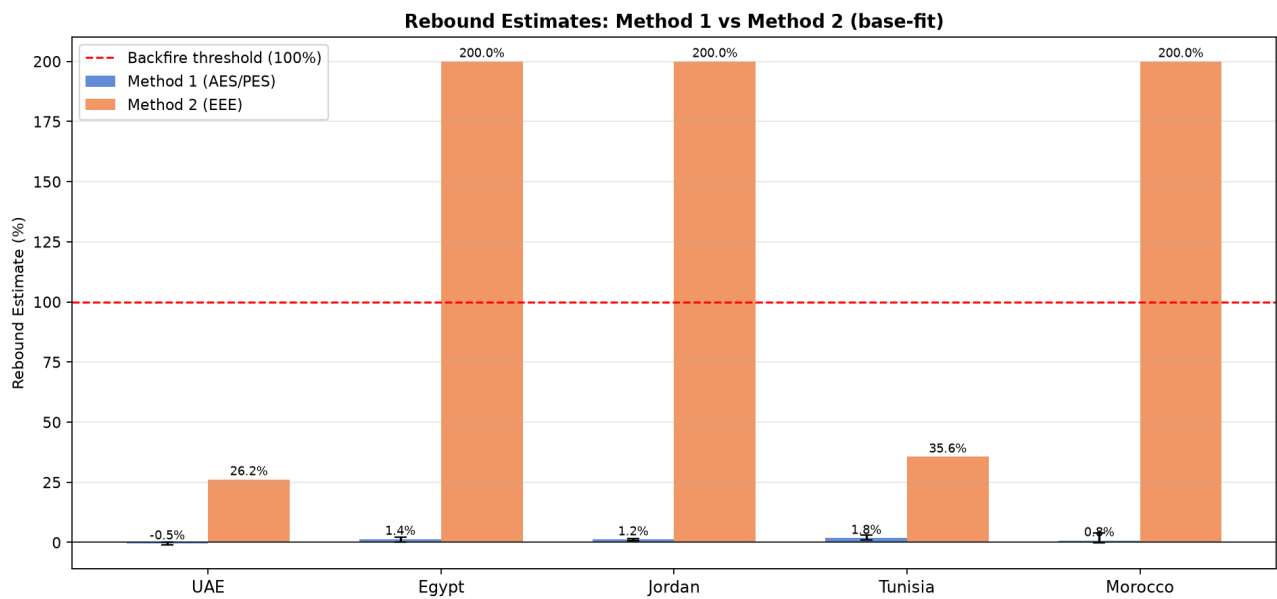


Figure 7. Grouped bar chart comparing Re_M1 (with 95% CI error bars) and Re_M2 base-fit for all five countries. Red dashed line = backfire threshold (100%).

Country	Re_M1 2.5%	Re_M1 Base	Re_M1 97.5%	Re_M2 2.5%	Re_M2 Base	Re_M2 97.5%	$\sigma(\text{KL-E})$	State M1	State M2
UAE	-0.90%	-0.50%	0.25%	200.00%	26.19%	26.19%	0.1667	Super-conservation	Partial rebound
Egypt	0.22%	1.40%	2.13%	200.00%	200.00%	26.19%	2.0000	Partial rebound	Backfire
Jordan	0.73%	1.18%	1.57%	200.00%	200.00%	36.84%	2.0000	Partial rebound	Backfire
Tunisia	0.91%	1.77%	3.00%	200.00%	35.61%	26.19%	0.2660	Partial rebound	Partial rebound
Morocco	-0.15%	0.82%	3.83%	200.00%	200.00%	27.59%	2.0000	Partial rebound	Backfire

Table 6. Comprehensive rebound summary for all five MENA countries. Colour-coded states: ■ Super-conservation ■ Partial rebound ■ Backfire.

9. Discussion

9.1 Method 1 (AES/PES) Findings

Method 1 yields low rebound estimates for all five countries ($< 2\%$), suggesting that efficiency improvements in MENA economies have generally delivered net energy savings. **UAE** is the single outlier showing *super-conservation* ($\text{Re_M1} = -0.50\%$), driven by a **negative** λ (-0.0046), implying slight embodied technological regress in the fitted CES — possibly reflecting the UAE's energy-intensive economic structure (oil sector dominance) partially offsetting efficiency gains. Egypt, Jordan, Tunisia, and Morocco all exhibit positive but small partial rebound ($0.8\%–1.8\%$), consistent with developing-economy patterns where income growth moderately offsets efficiency savings.

9.2 Method 2 (EEE) Findings

Method 2 results diverge substantially from Method 1. **Egypt, Jordan, and Morocco** register backfire ($\text{Re_M2} = 200\%$), a consequence of the optimizer converging to the lower bound of ρ (-0.5), implying $\sigma = 2.0$ (strong KL-Energy substitutability). Under such high substitutability, the EEE formula mechanically produces large rebound. **Tunisia** ($\text{Re_M2} = 35.6\%$, $\sigma = 0.27$) and **UAE** ($\text{Re_M2} = 26.2\%$, $\sigma = 0.17$) produce more moderate estimates, as their fitted ρ lies in the interior, implying complementarity between KL aggregate and energy.

9.3 Methodological Divergence

The large gap between Method 1 and Method 2 estimates is consistent with findings in Brockway et al. (2017), who note that the two methods measure different aspects of rebound: Method 1 captures observed energy intensity trends and attributes changes to technical progress (λ), while Method 2 derives rebound from structural elasticities (ρ) and is therefore highly sensitive to the CES parameter estimates. The boundary solutions for ρ in three countries limit the reliability of Method 2 for those cases, and suggest a need for richer data (more years, auxiliary instruments) to identify ρ precisely.

9.4 Limitations

- Energy proxy: using primary energy (Mtoe) as useful exergy proxy introduces measurement error — ideally, exergy conversion factors should be applied.
- Short CES panel (34 years) limits identification of ρ , particularly when growth in K, L, and E are co-linear.
- Uniform $\text{sE} = 0.08$ does not account for cross-country variation in energy prices and shares.
- Bootstrap CI for Method 2 is dominated by ρ uncertainty, producing very wide intervals when ρ hits its bound.
- λ is assumed time-invariant; in reality, embodied technical change may accelerate or decelerate.

10. Conclusions

Country	State M1	State M2	Key Finding
UAE	Super-conservation (M1)	Partial rebound (M2)	Negative λ , low σ (0.17) → energy-KL complementarity
Egypt	Partial rebound (M1)	Backfire (M2)	Moderate M1, but ρ at bound → M2 unreliable
Jordan	Partial rebound (M1)	Backfire (M2)	Moderate M1, but ρ at bound → M2 unreliable
Tunisia	Partial rebound (M1)	Partial rebound (M2)	Both methods agree → moderate rebound ~2% / ~36%
Morocco	Partial rebound (M1)	Backfire (M2)	Low M1 (0.8%), but ρ at bound → M2 unreliable

Overall, the AES/PES method suggests that energy efficiency policies in all five MENA countries have delivered genuine energy savings with only small rebound (< 2%). The EEE method produces higher and more uncertain estimates, driven by the structural elasticity of substitution. Tunisia stands out as the only country where both methods independently confirm partial rebound, providing greater confidence in its result. For Egypt, Jordan, and Morocco, the EEE-based backfire finding should be treated with caution due to boundary convergence of ρ .

11. References

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